Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
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Abstract

1. Problem. Student treasurers must submit an activity accountability report (LPJ) after funded campus events before subsequent funds are disbursed. Faculty guidelines enforce disparate administrative rules: receipt seals, arithmetic balance, attendance ratios, revenue stamps, tax deductions, and deadlines. At Brawijaya Computer Science, incomplete reports halt upcoming proposals and extracurricular peer learning [5]. Treasurers lose valuable study hours to mechanical bookkeeping corrections.

2. Intended Users. Treasurers and secretaries of Indonesian student organizations—study program associations, student executive boards (BEM), and clubs at Universitas Atma Jaya Makassar and peer institutions—who manage budgets without accounting training.

3. Supporting Observations or Evidence. Faculty research at Universitas Atma Jaya Makassar documented that the Student Affairs Bureau (BKAM) struggles with unstandardized reports and verification delays that threaten compliance [1]. Brawijaya Cultural Studies mandates stamped receipts, attendance lists per meal, and exact ledger balances [2]. UGM Economics enforces Rp10,000 revenue stamps on receipts of Rp5,000,000+ under Law 10/2020 and photocopies for thermal slips [3]. ITS Interior Design enforces a 10-day deadline [4], while ITS Biology requires separate line items and tax inclusion [5]. Student officers consistently struggle with tax classification and budget reconciliations [6].

4. Proposed Solution and Core User Flow. RapiLPJ is an agentic neuro-symbolic audit pipeline uniting Gemini multimodal perception with a deterministic verification engine. The flow has four stages: (1) Ingestion Agent (Gemini Flash Multimodal) extracts structured line items, dates, amounts, and visual compliance markers (ink seals, Rp10,000 meterai, thermal paper) from receipt photos. (2) Policy Agent extracts statutory thresholds from uploaded campus bylaws. (3) Deterministic Kernel executes zero-tolerance checks: duplicate invoices, date sanity, revenue stamp thresholds, and budget variances. (4) Remediation Co-Pilot Agent drafts official addendum letters. Treasurers export a cryptographic SHA-256 audit attestation sheet.

5. How It Differs from Existing Alternatives. Pure LLMs suffer from arithmetic hallucinations on financial tables. Spreadsheets lack multimodal ingestion, relational validation, and statutory bylaw parsing. Campus portals only accept documents post-hoc without pre-submission triage. RapiLPJ bridges this gap by coupling Gemini multimodal perception with a sandboxed deterministic audit engine, providing zero-hallucination verification, automated receipt digitization, and verifiable audit attestations.

6. Planned Technologies. A neuro-symbolic stack. Perception & Reasoning: Google Gemini 1.5/2.0 Flash Multimodal API with structured JSON outputs (response_schema) for receipt extraction, bylaw PDF compilation, and remediation drafting. Verification Kernel: client-side deterministic JavaScript engine executing statutory rules and Web Crypto SHA-256 attestation. Interface: responsive HTML5, CSS3 design tokens, and local storage, guaranteeing financial confidentiality.

7. Expected Benefit and Development Plan. Expected benefit: eliminates manual transcription, reduces LPJ revision cycles from 3–5 iterations to a single pass, and trains student leaders in statutory financial governance. Plan: Week 1 multimodal receipt ingestion trials with 30 treasurers at Universitas Atma Jaya Makassar; Week 2 bylaw PDF parser benchmarking across 5 faculties; Week 3 progressive web app refinement and printable attestation exports. OCR ambiguity is mitigated through deterministic confidence gating.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119. [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026. AI disclosure: Claude and Gemini supported research, deterministic engine code, and drafting.